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3.8 Let X be a nonnegative random variable on some probability space.

(a) Show that $(EX)(E\frac{1}{X}) \geq 1$. What does this say about the correlation between X and $\frac{1}{X}$?

(b) Let $f, g: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be Borel measurable and such that $f(x)g(x) \geq 1$ for all x in $\mathbb{R}_+$. Show that $Ef(X)Eg(X) \geq 1$.

(a) Suppose $X \geq 0$ a.e. (P)

By Cauchy-Schwarz inequality, $E(X)E(\frac{1}{X}) \geq \{E(X^{\frac{1}{2}} \cdot (\frac{1}{X})^{\frac{1}{2}})\}^2 = 1$

$\Rightarrow X$ and $\frac{1}{X}$ are always negatively correlated.

(b) $f(x)g(x) \geq 1 \Rightarrow f(x) \geq \frac{1}{g(x)}$ a.e. (P)

$$\Rightarrow E(f(X))E(g(X)) \geq E(\frac{1}{g(X)})E(g(X)) \geq 1$$

3.9 prove Corollary 3.1.3 using Hölder's Inequality applied to an appropriate measure space.

(1) Suppose $\mathcal{X} = \{1, \dots, k\}$, $\mathcal{F} = \mathcal{P}(\mathcal{X})$, μ is a probability measure st. $\mu(\{i\}) = \frac{C_i}{\sum_{i=1}^k C_i}$

$f(i) = a_i, g(i) = b_i, i=1, \dots, k. f(x)=g(x)=0$ if $x \notin \{1, 2, \dots, k\}$

$\Rightarrow f \in L^p(\mathcal{X}, \mathcal{F}, \mu)$ and $g \in L^q(\mathcal{X}, \mathcal{F}, \mu)$

By Thm 3.1.11 $\sum_{i=1}^k |a_i b_i| \frac{C_i}{\sum_{i=1}^k C_i} \leq \left(\sum_{i=1}^k |a_i|^p \frac{C_i}{\sum_{i=1}^k C_i} \right)^{\frac{1}{p}} \left(\sum_{i=1}^k |b_i|^q \frac{C_i}{\sum_{i=1}^k C_i} \right)^{\frac{1}{q}}. \quad q = \frac{p-1}{p}$

$$\Rightarrow \sum_{i=1}^k |a_i b_i| C_i \leq \left(\sum_{i=1}^k |a_i|^p C_i \right)^{\frac{1}{p}} \left(\sum_{i=1}^k |b_i|^q C_i \right)^{\frac{1}{q}}.$$

(2) Let $p=q=2$. proved.

3.17 Let X be a random variable on a probability space $(\mathcal{X}, \mathcal{F}, \mu)$. Recall that $Eh(X) = \int h(X) d\mu$ if $h(X) \in L^1(\mathcal{X}, \mathcal{F}, \mu)$

(a) show that $(E|X|^{p_1}) \leq (E|X|^{p_2})^{\frac{p_1}{p_2}}$ for any $0 < p_1 < p_2 < \infty$.

(b) show that " $\leq$ " holds in (a) iff $|X|$ is a constant a.e. (μ)

(a) Let $p = p_2/p_1 \geq 1, \frac{1}{p} + \frac{1}{q} = 1, Y = |X|^{p_1}$

By Hölder's inequality $E|X|^{p_1} = E(Y \cdot 1) \leq (E(Y^p))^{\frac{1}{p}} (E(1^q))^{\frac{1}{q}} = (E(Y^p))^{\frac{1}{p}} = (E|X|^{p_2})^{\frac{p_1}{p_2}}$

(b) '=' holds i.f.f. $Y = c$ a.e. (μ) i.e. $|X|$ is a constant a.e. (μ)

3.18 Let X be a non-negative random variable

(a) Show that $EX \log X \geq (EX)(E \log X)$

(b) Show that $\sqrt{1 + (EX)^2} \leq E(\sqrt{1 + X^2}) \leq 1 + EX$

(a) $f(x) = x \log x$ is convex in $(0, \infty)$

By Jensen's inequality $E f(X) \geq f(EX) \Rightarrow E(X \log X) \geq EX \log(EX)$

(b) Let $f(x) = \sqrt{1 + x^2} \Rightarrow f(x)$ is convex.

So $\sqrt{1 + (EX)^2} = f(EX) \leq E f(X) = E \sqrt{1 + X^2}$ if $EX < \infty$.

If $EX = \infty$, $\infty \leq \infty$ is true.

Also $1 + X^2 \leq (1 + X)^2$ so $E \sqrt{1 + X^2} \leq E(1 + X) = 1 + EX$.